

Vaishnavi Shrivastava

CONTACT	vaish.shrivastava@gmail.com · vshrivas.github.io · Google Scholar · LinkedIn		
EDUCATION	Stanford University	2022–2024	
	Master of Science, Computer Science Advisor: Percy Liang	3.9/4.0	
	California Institute of Technology (Caltech)	2015–2019	
	Bachelor of Science, Computer Science	3.9/4.0	
EMPLOYMENT	• Microsoft Research , Redmond, WA – Senior Researcher	Sep’24 – Present	
	Reinforcement learning · computer-use agents · reasoning		
	• Stanford University , Stanford, CA – Research Assistant	Sep’22 – Jun’24	
	Advisor: Percy Liang LLM calibration · self-play		
	• Allen Institute for AI , Seattle, WA – Research Assistant	Jun’23 – Dec’23	
PEER-REVIEWED PUBLICATIONS	Advisor: Tushar Khot Reasoning · LLM safety		
	• Microsoft AI , Bellevue, WA – Applied Scientist	Sep’19 – Aug’22	
	Efficient LLMs · personalization		
	• Software Engineering Internships – Microsoft, Dell EMC	2016–2018	
PEER-REVIEWED PUBLICATIONS	7. <u>ECHO: Terminal Agents Learn World Models for Free.</u>		
	V. Shrivastava, P. Kauffmann, A. Awadallah, D. Papailiopoulos. 2026.		
	Conference on Neural Information Processing Systems (<i>NeurIPS</i>), 2026 – <i>Spotlight</i>		
	6. <u>Wait, Wait, Wait... Why Do Reasoning Models Loop?</u>		
	C. Pipis, S. Garg, V. Kontonis, V. Shrivastava, A. Krishnamurthy, D. Papailiopoulos. 2025.		
	International Conference on Machine Learning (<i>ICML</i>), 2026 – <i>Spotlight</i>		
	5. <u>Sample More to Think Less: Group Filtered Policy Optimization for Concise Reasoning.</u>		
	V. Shrivastava, A. H. Awadallah, V. Balachandran, S. Garg, H. Behl, D. Papailiopoulos. 2025.		
	International Conference on Learning Representations (<i>ICLR</i>), 2026		
	4. <u>Benchmarking and Improving Generator-Validator Consistency of Language Models.</u>		
PREPRINTS & TECHNICAL REPORTS	X. Lisa Li, V. Shrivastava, S. Li, T. Hashimoto, P. Liang. 2023.		
	International Conference on Learning Representations (<i>ICLR</i>), 2024		
	3. <u>Bias Runs Deep: Implicit Reasoning Biases in Persona-Assigned LLMs.</u>		
	S. Gupta, V. Shrivastava, A. Deshpande, A. Kalyan, P. Clark, A. Sabharwal, T. Khot. 2023.		
	International Conference on Learning Representations (<i>ICLR</i>), 2024		
	2. <u>UserIdentifier: Implicit User Representations for Simple and Effective Personalized Sentiment Analysis.</u>		
	F. Miresghallah, V. Shrivastava, M. Shokouhi, T. Berg-Kirkpatrick, R. Sim, D. Dimitriadis. 2021.		
	North American Chapter of the Association for Computational Linguistics (<i>NAACL</i>), 2022		
	U.S. Patent 12,182,511, granted 2024		
	1. <u>Not All Thoughts Matter: Selective Attention for Efficient Reasoning.</u>		
PREPRINTS & TECHNICAL REPORTS	H. Tang, G. Zheng, K. Gandhi, H. Behl, V. Shrivastava, M. Javaheripi, K. Ellis, S. Garg, D. Papailiopoulos. 2025.		
	NeurIPS Workshop on Efficient Reasoning, 2025		
PREPRINTS & TECHNICAL REPORTS	5. <u>SuperThoughts: Reasoning Tokens in Superposition.</u>		
	Z. Xiong, S. Garg, M. Yu, V. Shrivastava, H. Zhao, A. Kyrillidis, D. Papailiopoulos. 2026.		
	arXiv preprint, 2026		

4. Phi-4-reasoning Technical Report.
M. Abdin, S. Agarwal, A. Awadallah, V. Balachandran, H. Behl, L. Chen, G. de Rosa, S. Gunasekar, M. Javaheripi, N. Joshi, P. Kauffmann, Y. Lara, C. C. T. Mendes, A. Mitra, B. Nushi, D. Papailiopoulos, O. Saarikivi, S. Shah, **V. Shrivastava**, V. Vineet, Y. Wu, S. Yousefi, G. Zheng. 2025.
Authors listed alphabetically by last name.
arXiv preprint, 2025
3. Language Models Prefer What They Know: Relative Confidence Estimation via Confidence Preferences.
V. Shrivastava, A. Kumar, P. Liang. 2025.
arXiv preprint, 2025
2. Llamas Know What GPTs Don't Show: Surrogate Models for Confidence Estimation.
V. Shrivastava, P. Liang, A. Kumar. 2023.
arXiv preprint, 2023
1. Exploring Low-Cost Transformer Model Compression for Large-Scale Commercial Reply Suggestions.
V. Shrivastava*, R. Gaonkar*, S. Gupta*, A. Jha. 2021.
*Equal Contribution
Microsoft Journal of Applied Research, 2021

TEACHING
EXPERIENCE

Teaching Assistant:

- **Caltech:** Machine Learning & Data Mining, CS 155 *Jan'19 – Mar'19*
- **Caltech:** Database System Implementation, CS 122 *Jan'18 – Mar'18*

REFERENCES

Dimitris Papailiopoulos, *Research Manager, Microsoft; Associate Professor, University of Wisconsin*
Ahmed Awadallah, *Partner Research Manager, Microsoft*
Percy Liang, *Associate Professor, Stanford University*
Milad Shokouhi, *Distinguished Engineer, LinkedIn*